

Problem

- Circadian rhythms dictate many of our biological processes.
- Chronobiology is the field of study regarding circadian rhythms and timing keeping systems
- Research is currently being undergone to identify disruptors of the rhythm to further understand human health concerns.
- Circadian rhythm takes place in brain; specifically the master clock in vertebrates.

Introduction

- Biological factors (biofactors) are made up of molecules produced or modified by living organisms which are present in many essential systems of an organism's body.
- Cells, tissues, and organs involved in metabolism are affected by circadian loops, driven by polar metabolites, such as lipids, that initiate change in circadian rhythms, ultimately influencing metabolism and aging of the body.
- Humans' moods and mental states of well being are controlled by hormones released by monoamines within the brain. The neurotransmitters released, such as serotonin, dopamine, endorphins, and norepinephrine, are all responsible for regulation of mood and emotional feeling.

Objective / Hypothesis

Objective: Is there a significant association between the daily shifts in a human's circadian rhythm vs. their metabolic rate and mood?

Hypothesis: There is a significant relationship between the daily shifts in humans' circadian rhythms to their metabolism (metabolic rate) and mood (melatonin levels).

Independent Variable: Fluctuations in circadian rhythms.

Dependent Variable: Changes in the melontin and metabolic rates.

Constants: Other factors such as brainly functions and body mechanisms.

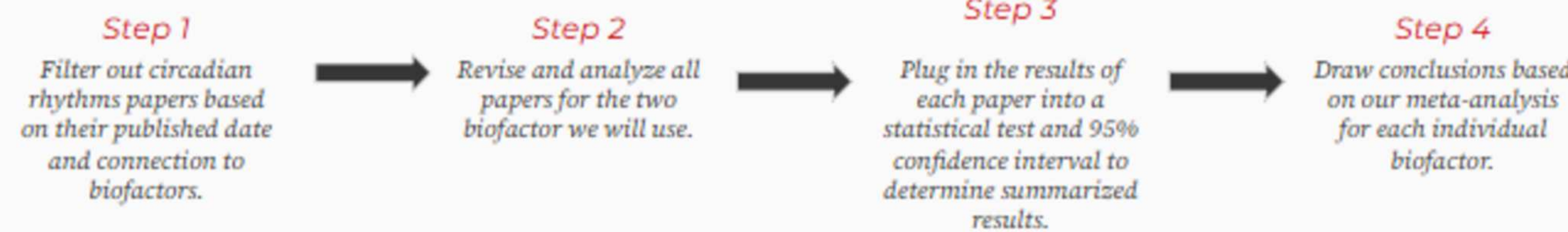
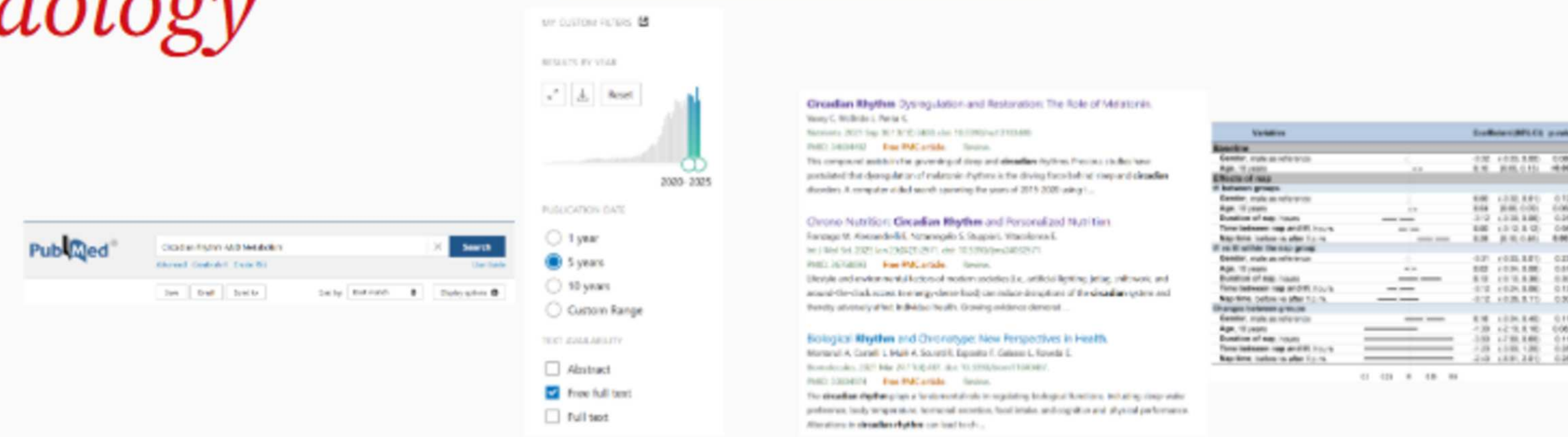
Controls: No controls.

A Meta-Analysis Between Humans' Biofactors and Circadian Rhythms

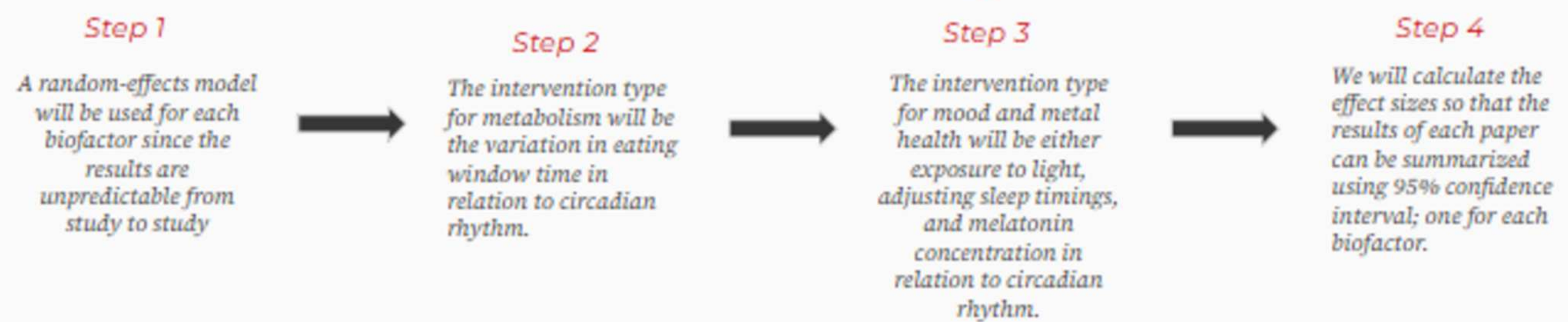
Methodology

Methodology

Filtration Procedure:



Statistical Testing:



$$d = \frac{\text{Mean}(\text{intervention}) - \text{Mean}(\text{control})}{\text{Pooled SD}}$$

Data / Results

	Country	Design	Intervention	Comparator	DurationWeeks	Outcome	Mean_Intervention	SD_Intervention	N_Intervention	Mean_Control	SD_Control
20	United States	Randomized crossover, controlled feeding	Prolonged daytime eating window (early/extended daytime meals)	Delayed eating window (late meals)	~2 (per condition; multi-day protocols)	Weight, fasting glucose, insulin sensitivity, circadian phase markers	91.6		19 (participants; crossover)	97	12
21	United States	Mouse study (~60 mice); Human correlation (20 older women)	Protein intake timing (morning vs evening meals)		8 weeks (mouse); cross-sectional (human)	Skeletal muscle hypertrophy (fiber cross-sectional area); muscle clock gene expression; breakfast protein intake correlation					
24	Italy	Randomized controlled trial	Early time-restricted carbohydrate consumption (eTRC)	Mediterranean-style diet	12	HbA1c (%)	-0.2	[convert from IQR: -0.5 to 0]	12	-0.3	[convert from IQR: -0.5 to -0.1]
24	Germany	Randomized controlled trial	Formula C (GOS supplementation)	Formula A (placebo)	24	Bifidobacterium abundance at 3 months	46	[not reported]	36	32.3	[not reported]
22	Sweden	Randomized crossover trial	Morning high-intensity interval training (08:00)	Afternoon high-intensity interval training (16:45)	2 (per arm, with washout)	Fasting glucose	7.7	0.9	5	7.6	
22	Netherlands	Randomized crossover trial	Time-restricted eating (10 h daily window, last meal ≤18:00)	Control (≥14 h eating window)	3	Fasting glucose (day 20)	7.6	0.4	14	8.6	0.4
22	Norway	Randomized controlled trial (4 arms)	Time-restricted eating (≤10 h/day)	Control (no intervention)	7	HbA1c (mmol/mol)	34.7	3.1	26	34.2	2.5
25	Germany	Randomized controlled crossover (single-blind)	6.5 h ambient hypoxia (FIO ₂ ~15%) during early night	Normoxia (FIO ₂ ~21%)	~1.5 (11-day protocol per arm)	Dim-light melatonin onset (DLMO) phase shift	Advance of 9 min	13.9 (SD from interim analysis)	21	Delay of 24 min	—
23	USA	Randomized controlled trial (CALERIE 2)	Calorie restriction (~12% sustained over 2 years)	Ad libitum diet	104	Muscle quality (mass-adjusted strength, gene expression)	Improved muscle quality despite lean mass loss; 1003 genes differentially expressed	—	57	Stable	—
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For each study, the following were recorded:

- Study ID
- Authors
- Year of publication
- Country
- Design
- Intervention
- comparator
- Duration
- Outcome
- Mean intervention
- SD intervention
- Mean Control
- SD Control
- Units
- Scale
- Bias risks

Discussion / Stats

The studies included in this meta-analysis show moderate heterogeneity in both methodology and outcome measures related to circadian disruption.

Variability in how circadian rhythm disturbances and outcomes like metabolism and mood are defined introduces challenges in direct comparison across studies.

Effect sizes such as Cohen's d or Hedges' g can be used for continuous variables, while odds ratios may apply to categorical outcomes.

Heterogeneity are to be assessed using I² statistics and Cochran's Q test to determine the consistency of results across studies.

Preliminary findings suggest circadian misalignment is associated with disruptions in metabolic processes and mood-related hormones, though inconsistencies in measurement limit strong conclusions.

Future research should prioritize standardizing definitions and selecting consistent biomarkers such as melatonin, cortisol, and glucose indicators.

All images that are not otherwise cited were created by student researcher.

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Berman, N. G., & Parker, R. A. (2002). Meta-analysis: Neither quick nor easy. BMC Medical Research Methodology, 2(10). <https://doi.org/10.1186/1471-2288-2-10>

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